

(ADAM) CHEOL WOO KIM

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ACADEMIC APPOINTMENTS

Harvard University

Postdoctoral Fellow in Computer Science

- Advisor: Milind Tambe

Boston, MA

Sep 2024 - Present

EDUCATION

Massachusetts Institute of Technology

PhD in Operations Research ; GPA: 5.0/5.0

- Advisor: Dimitris Bertsimas

- Thesis: Predictive and Prescriptive Trees for Optimization and Control Problems

- Coursework: Dynamic Programming and Reinforcement Learning, Online Learning and Optimization, Statistical Learning Theory and Applications, High-Dimensional Statistics, Probability Theory, Nonlinear Optimization, Robust Optimization, Linear and Discrete Optimization

Cambridge, MA

Aug 2024

Seoul National University School of Law

J.D. Candidate (withdrew to pursue PhD at MIT)

Seoul, Republic of Korea

Feb 2019 - Jul 2019

Seoul National University

B.A. in Economics (Summa Cum Laude); Major GPA: 4.24/4.30

Seoul, Republic of Korea

Feb 2019

- Concentration in Mathematical Economics and Microeconomic Theory

- Selected Coursework: Probability Theory, Real Analysis(I/II)(II/II), Discrete Mathematics, Differential Equations, Numerical Analysis, Mathematical Statistics, Econometrics, Linear Algebra(I/II)(II/II), Contract Theory, Microeconomics, Experimental Economics, Mathematics for Economics, Statistics for Economics

RESEARCH INTERESTS

Reinforcement Learning, RLHF, AI Alignment, ML-Accelerated Optimization

PAPERS

*: alphabetical or co-first author

Preprints

- Bertsimas, D.*, **Kim, C.W.*** & Niño-Mora, J*. (2024). *Optimal Control of Fluid Restless Multi-armed Bandits: A Machine Learning Approach*, Major Revision at Machine Learning
- Bertsimas, D.* & **Kim, C.W.*** (2024). *Optimal Control of Multiclass Fluid Queueing Networks: A Machine Learning Approach*, Revise and resubmit at Operations Research

Conference Publications

- Kong, L., Wang, H., Pan, Y., **Kim, C.W.**, Song, M., Nguyen, A., Wang, T., Xu, H., Tambe, M. (2025). *Robust Optimization with Diffusion Models for Green Security*, Uncertainty in Artificial Intelligence (UAI)
- **Kim, C.W.***, Moondra, J.*, Verma, S., Pollack, M., Kong, L., Tambe, M., Gupta, S. (2025). *Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning*, International Conference on Machine Learning (ICML)

Journal Publications

- Bertsimas, D.* & **Kim, C.W.*** (2024). *A Machine Learning Approach to Two-stage Adaptive Robust Optimization*, European Journal of Operational Research
- Bertsimas, D.* & **Kim, C.W.*** (2023). *A Prescriptive Machine Learning Approach to Mixed Integer Convex Optimization*, INFORMS Journal on Computing

RESEARCH & WORK EXPERIENCE

Massachusetts Institute of Technology

PhD Candidate

Cambridge, MA

Sep 2019 - Aug 2024

Core Thesis Topic

- Learning-accelerated algorithms for optimization and control.

Other Projects

- Developed machine learning models to predict patient flow at Beth Israel Deaconess Medical Center (a teaching hospital of Harvard Medical School), using medical data. The prediction models are fully integrated into their system.
- Collaborated with the OCP Group to develop a machine learning model to predict agricultural yield in Morocco combining image data and chemical composition data.
- Participated in a collaborative work to combat the spread of the Covid-19 pandemic:
(<https://www.covidanalytics.io/>)

Microsoft Research

Research Intern - Machine Learning and Optimization Group

Redmond, WA

May 2023 - Aug 2023

- Research Topic: Theory and algorithms for combinatorial optimization under uncertainty.
- Mentors: Ishai Menache, Marco Molinaro, Konstantina Mellou

Permanent Mission of the Republic of Korea to the United Nations

Research Intern - UN General Assembly Group

New York, NY

Jan 2014 - Apr 2014

- Research Topic: Revitalization of the UN General Assembly and its implications on Korea.

TEACHING EXPERIENCE

Developed course materials, led recitations, and held office hours for the following courses:

TA, The Analytics Edge (MIT 15.071)

Spring 2024

- MBA course on statistics, machine learning, optimization, and data science.

TA, Optimization Methods (MIT 6.7200J/15.093J)

Fall 2022

- Graduate course on linear, nonlinear, combinatorial, robust, and semi-definite optimization.

TA, Introduction to Mathematical Programming (MIT 6.251J/15.081J)

Fall 2021

- PhD-level course on linear, nonlinear, discrete, combinatorial, robust, and semi-definite optimization.

TA, Robust Modeling, Optimization, and Computation (MIT 1.142J/15.094J) *Spring 2021*

- Graduate course on theory and applications of robust optimization.

REVIEWER SERVICES

Journal of Machine Learning Research, INFORMS Journal on Optimization, European Journal of Operational Research, Transportation Science

AWARDS AND HONORS

Kwanjeong Educational Fellowship

Sep 2019 – Present

- Merit-based PhD scholarship.

Eminence Scholarship

Department of Economics, Seoul National University

Aug 2017

Seoul, Republic of Korea

- Awarded to top 1% in the economics department.

SKILLS

- **Programming:** Python (PyTorch, Scikit-learn), Julia (JuMP, Gurobi, Mosek), R
- **Languages:** Korean (Native), English (Fluent)